CARDIFF UNIVERSITY
PRIFYSGOL CAERDYDD

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Academic Year: 2023/24

Examination Period: Spring

Module Code: MAT700

Examination Paper Title: Mathematical Methods for Data Mining

Duration: 2 hours

Please read the following information carefully:

Structure of Examination Paper:

- There are 4 pages including this page.
- There are **4** questions in total.
- There are no appendices.
- The maximum mark for the examination paper is 75 and the mark obtainable for a question or part of a question is shown in brackets alongside the question.

Instructions for completing the examination:

- Complete the front cover of any answer books used.
- This examination paper must be submitted to an Invigilator at the end of the examination.
- Answer **THREE** questions.
- Each question should be answered on a separate page.

You will be provided with / or allowed:

- **ONE** answer book.
- Squared graph paper.
- The **use of calculators** is **permitted** in this examination.
- The use of a translation dictionary between English or Welsh and another language, provided that it bears an appropriate departmental stamp, is permitted in this examination.
- The use of the student's own notes, up to **1 sheet (2 sides) of A4 paper**, is permitted in this examination.

1. (a) What are supervised learning and unsupervised learning? Give examples. [6]

   (b) Describe the most common mathematical framework for learning. [6]

   (c) Define the *softmax* function and the *cross entropy* between two probability distributions over the same underlying set of events. [4]

   (d) Explain Bonferroni's principle for Data Mining. [4]

   (e) In a class of 250 students, suppose that the birthdays are independent and uniformly distributed over the 365 days of the year. We are interested in the event that some students have their birthday on the same day. Calculate the expectation of the number of triples of students having the same birthday. [5]

2. (a) Define the TF×IDF score for words in a collection of documents. [3]

   (b) Suppose there is a repository of 256 documents, and a word w appears in 64 of them. In a particular document D, the maximum number of occurrences of a word is 45. What is the TFxIDF score for w in D if that word appears six times in D? [3]

   (c) Show that in a corpus of 1500 documents, any word in any document has a TF×IDF score less than 11. [3]

   (d) Prove that the *edit distance* is indeed a distance measure. [4]

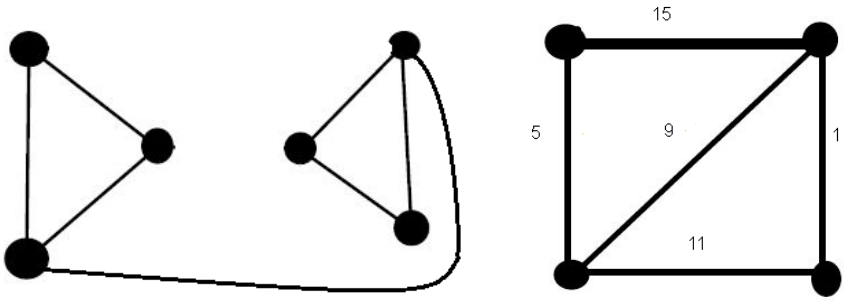
   (e) Compute the Jaccard similarities of the following two sets: $\{2, 3, 5, 7, 11, 13\}$ and $\{1, 3, 5, 7, 9, 11, 13, 15\}$. [2]

   (f) Compute the Jaccard similarities of the following two bags: $\{1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5\}$ and $\{5, 4, 4, 3, 3, 3, 2, 2, 2, 2, 1, 1, 1, 1, 1\}$. [2]

   (g) Prove that the Jaccard similarity of bags is always less than or equal to $1/2$. [4]

   (h) Define the Jaccard distance between sets. Prove that this function is a distance measure. [4]

3. (a) Find the adjacency matrices of the following graphs: [4]



(b) Let $G = (V, E)$ be an undirected graph. Show that [4]

$$\sum_{v \in V} \deg(v) = 2|E|.$$

(c) What is a meta-algorithm? Describe the halving algorithm and the weighted majority algorithm. [5]

(d) Compute the PageRank of each page in Fig. 1, assuming $\beta = 0.8$. [5]

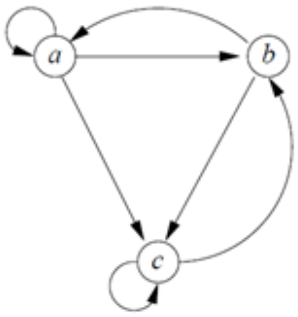


Fig. 1

(e) Describe how to calculate *precision*, *recall* and *F-measure* of a binary classifier. [3]

(f) Prove that F-measure of any binary classifier is $\leq \frac{\text{precision+recall}}{2}$. [4]

4. (a) Suppose we have access to a set of n experts and assume that one of the experts is consistent. Prove that the halving algorithm makes at most $\log_2 n$ mistakes. [5]

(b) Consider the one-dimensional data set shown in the Table

x	1	4	9	16	25	36	49	64	81
y	+	+	−	−	−	?	+	−	+

- Compute the class of x = 36 according to its 2-, 3-nearest neighbours (using the majority vote). [4]
- Repeat the previous task, but use the weighted version of kNN. [4]

(c) Assume we have the following data:

outlook	temperature	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

In this dataset, there are five categorical attributes *outlook*, *temperature*, *humidity*, *windy*, and *play*. We are interested in building a system to decide whether to play or not to play the game based on the weather conditions. Consider a new data instance X = *(sunny, cool, high, true)*. How would the Naïve Bayesian classifier classify X? [5]

(d) Formulate the k-means algorithm. What is the objective function of the k-means algorithm? Prove the convergence of the k-means algorithm. [7]